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The Phenomenological Stability of Depersonalization: Comparing the Old with the New

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The view that depersonalization is a stable syndrome became well established during the first half of the 20th century. Current operational definitions restrict depersonalization to the experience of unreality. This is likely to neglect clinical features of potential neurobiological relevance. By using the year 1946 as the dividing line, 200 cases of depersonalization disorder reported in the medical literature since 1898 were divided into two historical groups (1 and 2). The groups were then compared in terms of 18 phenomenological variables with a sample of 45 prospective cases of DSM-IV depersonalization disorder (group 3 or gold standard). Groups 1 and 2 differed in terms of their symptom profile, but the highest frequency that symptoms achieved in either group did not differ from the rates identified in group 3. A core of (invariable) symptoms, including emotional numbing, visual derealization, and altered body experience, was present throughout. These high rates of spontaneous reporting in all three groups may be explained by the fact that they all are accompanied by specific distress. With the exception of heightened self-observation and altered time experiencing, all other symptoms were significantly lower in group 2. The results suggest that the phenomenology of depersonalization has remained stable over the last 100 years. Our study found differences in frequency for some symptoms, but these are likely to have resulted from reporting biases, themselves governed by changing theoretical views. Clinical descriptions became poorer as the present is approached. This cannot be solely explained on the basis of empirical progress, and it is likely that theoretical biases also play a role. Because the neurobiological relevance of the symptoms of depersonalization remains unknown, it makes sense to continue collecting as many symptoms as possible, thereby avoiding both biased selection or premature closure.

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Depersonalization is defined by DSM-IV as "a feeling of detachment or estrangement from one's self. The individual may feel like an automaton or as if he or she is living in a dream or a movie. There may be a sensation of being an outside observer of one's mental processes, one's body, or parts of one's body. Various types of sensory anesthesia, lack of affective responses, and a sensation of lacking control of one's actions, including speech is often present" (p. 488, American Psychiatric Association, 1994).

Both the term and concept of depersonalization were constructed during the late 19th century (Sierra and Berrios, 1997). Derealization was only coined during the early 20th to refer to an aspect of depersonalization, namely, "an alteration in the perception or experience of the external world so that it seems strange or unreal (*e.g.*, people may seem unfamiliar or mechanical)" (American Psychiatric Association, 1994). Unless specifically indicated, depersonalization will be used here as the generic term for both aspects as there is not conclusive evidence that derealization is an independent phenomenon.

The Phenomenology of Depersonalization

The phenomenological richness that once characterized definitions of depersonalization has suffered somewhat on account of the fact that current operational definitions often depend on negative descriptors (*e.g.*, feelings of unreality). In addition to objects seeming different or unreal, patients also complain of emotional numbing (Ackner, 1954); heightened self-observation (Saperstein, 1949); changes in body experience (Cappon and Banks, 1965; L'hermitte, 1939); absence of bodily feelings such as hunger, thirst, etc. (Mayer-Gross, 1935); changes in the experiencing of time (Cappon and

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Banks, 1969) (Lewis, 1931) and space (Cappon, 1969; Shorvon, 1946); feelings of not being in control of movement (*i.e.*, loss of feelings of agency); having the mind empty of thoughts, memories, and images (Amabile and Rizzo, 1966); and inability to focus and sustain attention (Mayer-Gross, 1935; Shorvon, 1946). Sometimes, the feelings of unreality seem to affect one perceptual modality (*e.g.*, vision or audition), and in this sense they can be modality-specific (Sierra and Berrios, 1998). The mapping of depersonalization seems to remain an important challenge given that it is yet unclear which of the symptoms listed above (and others that may not have yet been described) are biologically significant. For example, even uncommonly reported symptoms, such as the experience of mind emptiness or dulled pain, may have neurobiological relevance (Sierra and Berrios, 1998; see Table 1 for a glossary of these terms).

As is the case with other neuropsychiatric phenomena (Berrios, 1996), the construction of depersonalization was based on single cases (Sierra and Berrios, 1996) and on the unsystematic description of small samples (Mayer-Gross, 1935; Shorvon, 1946; Roth, 1959). Studies based on larger samples mostly focus on demographic variables, natural history, comorbidity, etc., but leave aside the phenomenological (experiential) features of depersonalization (*e.g.*, Simeon et al., 1997). In spite of the above, depersonalization is likely to constitute a stable syndrome (Mayer-Gross, 1935; Saperstein, 1949; Shorvon, 1946).

This paper will map the symptom-profile of depersonalization (and changes thereof) since its original description and compare the clinical features of a large historical sample with a prospective cohort of patients suffering from DSM-IV depersonalization disorder.

Cases and Methodology

Historical Sample

Based on a comprehensive search of the literature in five European languages (sources: *Index Medicus*, MEDLINE, *Psychological Abstracts*, and a full search of relevant historical books on psychopathology), 147 clinical studies were identified containing cases explicitly diagnosed by their authors as depersonalization because this symptom was their single (or predominant) clinical manifestation. The publication date for these studies ranged from 1898 (*i.e.*, when the word and concept of depersonalization were first proposed) to 1996. The original cohort of 275 cases was pared down to 200 by excluding reports with: a) a poor phenomenological description (poor is defined here as a report that mentions less than 3 components of the syndrome), and b) insufficient information to satisfy the first three DSM-IV criteria for depersonalization disorder. (Criterion D, that "the experiences does not occur exclusively during the course of another mental disorder..." was *not* used as the objective of the study was to obtain as complete a clinical description of the depersonalization phenomenon as possible.) The exclusion criteria are unlikely to have biased the clinical profile of our database as a study of the excluded cases showed that they did not differ in age, gender, or background pathology from the index sample.

Clinical information was collected in a standard protocol recording 18 variables: age; gender; mental functions involved in the experience of unreality, including type of sensory modality involved; presence of emotional numbness; changes in the feeling of agency; distortions in the experiencing of time; changes in body experience; heightening of self-observation; feelings of thought emptiness or changes in the subjective experiencing of memory; and subjective feelings of an inability to evoke images (for operational definitions see Table 1).

Information was also recorded on autoscopy; out-of-body experiences; déjà vu (symptoms often reported in association with depersonalization). Each variable was endorsed as present if the description matched the corresponding operational definition. A symptom was declared absent when the original author reported either it thus or when it was not mentioned as present.

To detect changes in description, and because the most important work on the phenomenology of depersonalization was published before the Second World War (*e.g.*, Schilder, 1914; Haug, 1936; Mayer-Gross, 1935), the historical sample was divided into a phenomenological period (group 1 = 94) and an empirical period (group 2 = 106) using the year 1946 as the dividing line.

Prospective Sample

A sample of 45 patients was collected that met criteria for DSM-IV depersonalization disorder (group 3). All subjects were administered the Cambridge Depersonalization Scale (CDS; Sierra and Berrios, 2000), a comprehensive instrument whose 29 items include all the variables covered by the historical questionnaire used in the current study. Each item of the CDS includes two Likert scales (frequency and duration) and the sum of their scores is considered as a measure of overall item intensity. To be able to compare historical data (where variables were rated as present/absent) and the CDS,

TABLE 1
Glossary: The Components of Depersonalization

Symptom	Definition	Example
Emotional numbing	Absence or reduction in the experiencing of emotional feelings. In depersonalization the latter complaint occurs in the context of apparently normal motor emotional expression.	"He had unexplained crying episodes but he said not to be able to feel sadness nor happiness" (Haug, 1936).
Changes in body experience	Subjective feelings of body change regarding size of parts of the body, the feeling of being weightless, or the feeling as if parts of the body didn't belong to self.	"My hands do not seem to be mine . . .: The top part of my head often seems to disappear" (Roberts, 1960). "Generally my head and lower jaw feel disproportionately large for my body and my arms and fingers pencil thin" (Roberts, 1960).
Changes in visual experience	Any complaint of "unreality" concerning visual modality.	"Everything seems just a painted picture, deathly quiet" (Lewis, 1931). "Things looked dull, flat, and lifeless" (Lewis, 1931).
Changes in auditory experience	Complaints of unreality affecting the auditory modality	"Is there something wrong with my ears? I hear you clearly, yet your voice sounds far away, distant and unreal" (Bockner, 1947). "I could hear my voice and that of other people's very remotely although there was no lessening of sound" (Roberts, 1960).
Changes in tactile experience	Complaints of unreality affecting the haptic modality	["The patient complained that] things that he touched lacked vividness and reality" (Heuyer and Dublineau, 1932).
Changes in gustatory experience	Change in gustatory experiencing, so that the flavour of meals is no longer accompanied by feelings of pleasure or distaste.	"I taste but it does not mean anything to me. I'll eat anything put before me. We had walnut cake which I normally adore—but it might have been dry bread" (Bockner, 1947).
Changes in olfactory experience	Change in olfactory experience, so that smells are no longer accompanied by feelings of pleasure or dislike.	"Perfume doesn't smell pleasant any longer. I'd have no preference for the smell of these roses over the smell of cabbage cooking" (Bockner, 1947).
Loss of feelings of agency	The feeling that one is not in charge of one's movements or mental activity.	"It seemed rather pointless to reply to remarks made, but I heard my voice doing so" (Roberts, 1960). "I was walking and talking, as though automatically, I couldn't feel any movement and yet I knew I was walking" (Roberts, 1960).
Distortions in the experiencing of time	Subjective changes in the perception of duration (*i.e.*, time is experienced slower, faster or standstill), and in time perspective, *e.g.*, an inability to conceive past or future.	"Time doesn't seem to go at all; the whole day seems exactly the same. If I look at the clock it doesn't mean anything to me at all. I couldn't give you the remotest idea now what the time is [asked, said it was twelve o'clock, actually 11:50]. It's difficult to think of the past at all . . . everything seems to have gone out of my head" (Lewis, 1931).
Changes in the subjective experiencing of memory	The subjective feeling of not being able to recall things (*e.g.*, memory episodes) or having the feeling that the person was not part of the episode.	"When he recalled events in his life, he felt as though he was "not in them" (Fichter et al., 1992). "He had doubts as to whether his memories were of real events or of just a vivid dream" (Haug, 1936).
Feelings of thought emptiness	The feeling of having the mind empty of thoughts or not being able to think.	"Nothing comes into my head, there is an emptiness there. I don't seem to think unless anyone says anything to me" (Roberts, 1960). "I am not able to think, I don't have any thoughts, everything seems the same in my head, I try to think but I can't, my head is so weak, I can talk about anything, but nothing happens in my head" (Hartmann, 1922). "I can't collect my thoughts sufficiently. My thinking isn't clear" (Amabile and Rizzo, 1966).
Subjective feelings of an inability to evoke images	Complaints of inability to recall things as visual or auditory images.	"I can't picture things vividly or recall them properly" (Ackner, 1954). "He would constantly try to imagine his wife or his home, but he wouldn't succeed" (Hartmann, 1922).
Heightened self-observation	State of increased self-awareness in which one has the feeling of being a disembodied observer contemplating one's actions and mental activity.	"I seemed to be completely apart from myself. I felt that I was somewhere above looking down on the scene of which I was a part and yet not a part" (Roberts, 1960). "I was completely unable to tell whether I myself was still present or whether I was the part which had gone. In short there were two different beings, the one watching the other" (Roberts, 1960).

information in the later was rendered dichotomous thus: *absent* = never + rarely; and *present* = often + very often + all the time. Likewise, two outcome measures, one for emotional numbness and the other for altered body experience were created by collapsing the related four CDS and three CDS items, respectively. In this way, the scalar data from the CDS could be matched to the categorical data from the historical data sheet.

Statistical analysis was carried out by means of the SPSS (Statistical Package for the Social Sciences, Version 6.1). As recommended for dichoto-

TABLE 2
Symptom Distribution

Symptom	Pre 1946 (N = 94)	Post 1946 (N = 106)	Prospective (N = 45)	Kruskal Wallis Bonferroni-Corrected
Changes in visual experience	59 (62.8%)	61 (57.5%)	35 (77.8%)	NS
Changes in auditory experience	45 (47.9%)	14 (13.2%)	25 (55.6%)	p = .0014
Changes in tactile experience	17 (18.1%)	6 (5.7%)	11 (25%)	p = .047
Changes in gustatory experience	10 (10.6%)	6 (5.6%)	14 (30.6%)	p = .0042
Changes in olfactory experience	8 (8.5%)	2 (1.8%)	9 (19.5%)	p = .023
Emotional numbing	65 (69.1%)	64 (60.3%)	28 (62.2%)	NS
Changes in body experience	65 (69.9%)	65 (61.3%)	28 (62.2%)	NS
Changes in the subjective experience of memory	27 (29%)	23 (21.7%)	14 (31.1%)	NS
Loss of the feelings of agency	38 (40.8%)	34 (32%)	17 (37.7%)	NS
Feelings of thought emptiness	58 (62.3%)	31 (29.2%)	31 (68.8%)	p = .0001
Inability to evoke images	49 (52.6%)	20 (18.8%)	19 (42.2%)	p = .0014
Heightened self-observation	15 (16.1%)	36 (34.4%)	12 (26.7%)	p = .045
Distortions in the experiencing of time	15 (16.1%)	34 (32.4%)	14 (31.1%)	p = .044

mous data, the chi-squared statistic (with one degree of freedom and Bonferroni correction) was used throughout (Hall and Bird, 1986; Reynolds, 1984), unless otherwise indicated.

Results

Demographic Characteristics

Historical Sample. Mean age was as follows: group 1 = 30.24 (SD: 11.89); group 2 = 28.5 (SD 15.34); group 3 = 32.2 (SD: 7.3). Gender distribution was as follows: group 1 = male (43.6%), female (56.4%); group 2 = 47.2% and 52.8%; group 3 = 58.3% and 47.791%. There was no statistical difference across groups in terms of age or gender. The frequency of déjà-vu, out-of-body experiences, and autoscopy (phenomena allegedly concomitant of depersonalization) was very low (<5%), and hence these symptoms were excluded from further analysis. Table 2 shows symptom distribution across the three groups. The mean frequency of reported symptoms across groups was as follows: group 1 = 5.7 (SD 2.6); group 2 = 4 (SD = 2.3); and group 3 = 6.4 (SD = 3.9). The lower frequency in group 2 was statistically significant (chi-square = 22.1, df = 2, p = .0001).

The following symptoms occurred with similar frequency across the three groups: visual unreality, altered body experience, emotional numbing, and feelings of loss of agency (*i.e.*, automatization feelings) and had the highest frequencies overall (with the exception of feelings of loss of agency; see Table 1).

In each group, symptoms of unreality across modalities showed the same pattern of distribution. In each group, visual unreality achieved the highest frequency, followed by auditory, tactile, gustatory, and olfactory unreality. However, with the exception of visual unreality, all sensory modalities showed significantly lower frequencies in group 2 (Table 1). Differences for these variables (with the exception of the gustatory modality) between groups 1 and 3 were nonsignificant (audition, p = .43; touch, p = .395; and olfaction, p = .08), making it clear that the significant differences for the three groups reported in Table 1 are due to the low frequencies in group 2. Gustatory modality frequencies in group 3 were significantly higher than in group 1 (chi-square = 20.2, df = 2, p = .0065).

For Group 2, subjective absence of thoughts and inability to evoke images showed significantly lower differences (pairwise differences between groups 1 and 3 were not significant: absence of thoughts, p = .412; inability to evoke images, p = .287). The opposite pattern is seen in group 1, in which altered time experience and self-observation had significantly lower frequencies (pairwise comparison between groups 2 and 3 showed no significant differences: altered time experience, p = .631; heightened self-observation, p = .615)

Discussion

Analysis of historical phenomenological databases has been successfully used in the mapping of disorders such as autoscopy (Dening and Berrios, 1994), dysmorphophobia (Berrios and Kan, 1996), Cotard's syndrome (Berrios and Luque, 1995), suicide (Seidel, 1995), and the psychiatric manifestations of Wilson's disease (Dening and Berrios, 1989). Like any other data capturing method, it has advantages and disadvantages: a) it allows for the capture of samples larger than what is ordinarily available in prospective clinical studies; and b) it can detect shifts in symptom-patterns over time, providing information on the stability (or otherwise) of psychopathological constructs. The limitations of the

method are also well known, and they are similar to those of other meta-analytical techniques, namely, that publication events are not random but at the mercy of biases that tend to favor the positive and unusual. This is not, however, fatal as biases are likely to be two-tailed, and this seems to have been the case in our data. Because cases come from different diagnostic epochs, it could also be argued that they do not refer to the same disorder. However, because all historical cases met DSM-IV criteria for depersonalization disorder, this is unlikely to be the case. Thus, in spite of these limitations, the clinico-historical methodology (or statistical history) remains an economical way of exploring the phenomenology of any disorder, informing the validity of definitions and generating hypotheses for prospective research. To our knowledge, our paper is the first to compare a historical sample with a prospective sample of patients.

Descriptive Psychopathology of Depersonalization: Old and New

An interesting finding of our study was that earlier and current descriptions of depersonalization differ in *specific* ways. Rather than an artifact, this is likely to reflect differing modes of description or actual changes in the object under study. To explore this further, we used the prospective sample as our gold standard (group 3). The fact that in the symptom comparison between groups 1 and 2, the highest frequency of each pair was similar to that shown by group 3 suggests that the syndrome of depersonalization has remained stable, and that all differences obtained were due to underreporting for groups 1 or 2. With the exception of heightened self-observation and altered time experiencing, all symptoms in which differences were found had significantly lower frequencies in group 2, which supports the view that, in general, phenomenological descriptions of depersonalization have become impoverished. However, the opposite is seen in regard to self-observation and time experiencing. Our findings suggest that in earlier work these two symptoms were underreported and that their real frequency has been achieved (group 3). Causes for underreporting in the historical samples are likely to implicate both patients and clinicians. For example, the former may *spontaneously* declare only those symptoms that are most distressing. The converse is also the case and symptoms causing little or no distress may be underreported. We propose that being accompanied by a distress marker increases the chances for a symptom to be recorded by clinicians (capture of the other symptoms requires specific questioning). Symptoms affected by this bias would be expected to exhibit little variance across samples. This could explain the transhistorical or transcultural stability of some symptoms, which according to our findings are visual unreality, altered body experiencing, loss of agency, and emotional numbness. According to this hypothesis, derealization in the visual world (things lacking vividness, looking flat, etc.) tends to cause more distress and is often the subject of spontaneous complaints, whereas derealization affecting other sense modalities seems to cause less distress (as reported by patients) and hence is reported only after specific questioning. In keeping with this hypothesis are the findings by Shorvon (1946), who asked 66 patients with primary depersonalization *spontaneously* to describe their symptoms: "Individual cases describing some of these disturbances probably experience some other changes but omitted to mention them at interviews, particularly so because of the *care we have taken to avoid leading questions"* (our italics, Shorvon, 1946; p. 42). Although the actual figure is not provided, most of these patients seem to have complained of visual derealization; thus, the author commented: "there is a relative absence of olfactory, gustatory, or auditory derealization" (p. 791). The most frequent complaints seem to have been feelings of body change (68%) and loss of agency feelings (45%). As discussed above, these three symptoms belong to our group of invariable symptoms, and the frequencies we found are strikingly similar to his. It is also the case that symptoms with high spontaneous reporting (distress marker) are more likely to become enshrined in conceptual models of depersonalization.

The frequency of symptoms not volunteered by patients is dependent upon the keenness of the interviewer, and the latter in turn will be influenced by theoretical frames, known to change from one historical period to the next. Thus, symptoms affected by clinician-related biases may show high variance on historical analysis and create an illusion of unstability. Our study showed that to this category belong unreality experiences (in all modalities other than vision), the experience of mind emptiness (subjective inability to entertain thoughts or evoke images), heightened self-observation, and altered time experience. In this regard, Edwards and Angus (1972) have commented that the diagnosis of depersonalization "is not just a reflection of the skill of the interviewer, but is dependent on the type of questions and the manner in which they are asked. The less structured the interview, the more likely it is that there will be inconsistencies in the frequency with which depersonalization is recognized." It can

thus be concluded that not all the symptoms of depersonalization are equally susceptible to clinician-related biases. The frequency profile of symptoms pertaining to all sensory modalities (including the visual one) in group 1 is similar to that of group 3 (prospective sample) and thereby reflect the phenomenological complexity of depersonalization.

As we have suggested elsewhere, early accounts conceived of depersonalization as a disturbance of the *primary senses* (Sierra and Berrios, 1997). For example, Krishaber (1873), based on his observation of patients complaining of unreality feelings, took the view that these symptoms were caused by multiple sensory distortions and concluded that feelings, of unreality resulted from pathological changes in the sensory apparatus. Later on, this view was endorsed by Pick and Wernicke (Sierra and Berrios, 1997). Such theoretical notions in all probability encouraged earlier writers to overemphasize symptoms that seem to bear this out. In contrast, reports of unreality pertaining to sensory modalities other than vision drop dramatically during the second half of the 20th century. This is particularly striking in relation to the auditory modality whose frequency of reporting is similarly high in group 1 and in group 3 (the gold standard). This suggests the presence of a theoretical bias favoring vision; indeed, the very influential view put forward by Paul Schilder during the mid 1930s and 1940s is likely to have been a causal factor here (Schilder, 1935, 1953). Thus, in his seminal *The Image and Appearance of the Human Body*, he proposed a psychodynamic model of depersonalization emphasizing the visual modality and the denial of a voyeuristic tendency (Schilder, 1935, p. 140). He also suggested that heightened self-observation (once again related to the *eye metaphor*) was an important mechanism in the genesis of depersonalization. The wide acceptance of his model on both sides of the Atlantic could explain both the drop of complaints related to sensory modalities other than vision (as per group 2) and the increased rate of reports emphasizing heightened self-observation.

Frequency of gustatory complaints was significantly higher in group 3 (olfactory changes exhibit a similar trend, but the difference was not statistically significant). We believe that this may simply reflect differences in the way in which (as compared with vision and audition) smell and taste are used in current society (*i.e.*, intermittently and less frequently, at least at the conscious level). It is to be expected therefore that our questionnaire (CDS) might have picked up higher frequencies by specifically asking after such complaints which may not have been volunteered on the nonsystematic interviews carried out on the historical samples.

Post's (1946) descriptions (group 2) also show fewer cognitive complaints, such as subjective experience of absence of thoughts and inability to evoke images. These occur similarly and more prominently in groups 1 and 3. This may also be due to a theoretical bias. One of the important criticisms that had been levelled against the earlier sensory hypothesis of depersonalization was from Pierre Janet, who pointed out that it depended upon an obsolescent, passive view of the mind (on this, see Sierra and Berrios, 1997). Thus, the French psychologist wrote: "I have insisted on this essential point. A scientific psychology should consider psychological facts as actions and relate to them in those terms" (Janet, 1928; p. 101). In this regard, he believed that depersonalization concerned a *sentiment de vide* ("feeling of emptiness"), which he ascribed to a pathology of action: "the feeling of emptiness is a trouble of action, not of sensation or misunderstood consciousness" (Janet, 1928; p 101). For Janet, all psychical activity was either primary or secondary: the former encompassed all activity evoked by external stimuli—from knee jerks to evocation of memories; the latter was a background echo elicited by *representations* of primary acts. By conferring upon primary experiences a "feeling of vividness" (*I'impression de vie*), this secondary echo created the illusion of a continuous flow of psychic activity: "thousands of resonances constituted by secondary actions, fill the spirit during the intervals between external stimuli, and give the impression that it is never empty.... This complex activity links up the [primary] actions brutally determined by the external world, and causes an impression of vitality, or spontaneity, and at the same time of certitude" (Janet, 1928; p. 126). This new mechanism displaced the sensory hypothesis and its strong cognitive bias may well have sensitized clinicians to look for experiences phenomenologically germane such as mind emptiness. To test this hypothesis empirically, the prevalence of feelings of emptiness of thoughts during three periods, before, during, and after Janet's period of highest influence (1905–1930), was ascertained within group 1. Our findings seem to support the hypothesis as there were 6 cases (18%) in the first period, 35 (74%) in the second, and 48 (35.8%) in the third; these changes are statistically significant (Bonferroni corrected $p = .0001$ and $p = .004$, respectively). A similar fluctuation was observed in relation to memory complaints but did not attain statistical significance.

Along with heightened self-observation, reports of changes in the experience of time increased in group

2, reaching similar frequencies as those present in the prospective sample; once again, this suggests that some theoretical bias was in play for group 2. In fact, self-observation and altered time experiences are seldom mentioned in the early literature on depersonalization. In 1931, however, Lewis proposed that changes in the experiencing of time were not only an invariable finding, but also a central mechanism in depersonalization. This view was later expanded by Freeman and Melges (1977, 1978) and Winnik (1947–1948), who suggested that the experience of a time perspective provides a frame of self-reference without which the experience of self becomes distorted and unfamiliar. Indeed, recent studies have reported an association between temporal discontinuity and feelings of unreality, and it is assumed that time distortion is the primary experience (Freeman and Melges, 1977, 1978; Mathew et al., 1993). As with other types of experiences, it is likely that attention to the prominent role that the time experience may play has been instrumental in detecting it more frequently.

To sum up, the findings of our study support the hypothesis that the difference in the symptom-profile of depersonalization found in two successive historical periods may be partially explained by changes in concepts and explanatory views that emphasized different components of the disorder. This may have sensitized successive generations of clinicians to search for and report different symptom constellations. Far from being of purely academic interest, these findings should be considered as central to current endeavors to understand the neurobiological basis of mental symptoms (in general). The need for a complete phenomenological mapping of mental phenomena cannot be overemphasised, for there is no way a priori to tell which are likely to be of neurobiological relevance.

The idea that to save time it should suffice to select only a few salient clinical phenomena (salient being defined here as central to some theoretical view) is a risky strategy as this can lead to the neglect of mental phenomena that in the event will prove central to the biology of the condition under study. For example, the distinction made in this paper between visual and auditory feelings of unreality (neglected in official definitions) might turn out to have important neurobiological implications. Indeed, when analyzing the experiences of unreality reported by a series of patients with epilepsy (called illusions by the authors), Mullan and Penfield (1959) observed that visual unreality experiences occurred more frequently upon electrical stimulation of the right temporal lobe, whereas auditory unreality experiences occurred after stimulation to either side and no lateralization effect was observed. This kind of symptom-localization linkage needs to be pursued with more vigor.

Summary and Conclusions

The findings reported in this paper suggest that the phenomenology of depersonalization has remained stable over the last 100 years. In spite of this, the clinical profile of published case reports has been found to vary in specific ways. These variations are likely to have been determined by changing theoretical views on depersonalization. Guided by theory, clinicians were sensitized to perceive and report the same clinical phenomena in selective ways.

Two clusters of symptoms were identified. An invariant core included visual derealization, altered body experience, emotional numbing, loss of agency feelings, and changes in the subjective experiencing of memory; the distress caused by these complaints is likely to maximize spontaneous reporting. The second cluster includes symptoms whose reporting is unlikely to be spontaneous and whose capture, therefore, may be governed by theory. Two lessons can be drawn, one that there is no such a thing as an atheoretical psychopathological description; the other, that the identification of theoretical biases is essential to the understanding of both the structure and frequency of symptoms and for current neurobiological research on depersonalization (Cappon, 1961).

References

Ackner B (1954) Depersonalisation: I. Aetiology and phenomenology. *J Ment Sci* 100:838–853.

American Psychiatric Association (1994) *Diagnostic and statistical manual of mental disorders* (4th Ed). Washington, DC: Author.

Amabile G, Rizzo PA (1966) Perdita della visione mentale: (contributo clinico) *Rivista Sperimentale di Freniatria* 90:1156–1163.

Berrios GE (1996) *The history of mental symptoms*. Cambridge: Cambridge University Press.

Berrios GE, Kan CS (1996) A conceptual and quantitative analysis of 178 historical cases of dysmorphophobia. *Acta Psychiatr Scand* 94:1–7.

Berrios GE, Luque R (1995) Cotard's syndrome: Analysis of 100 cases. *Acta Psychiatr Scand* 91:185–188.

Bockner S (1947) The depersonalization syndrome: Report of a case. *J Ment Sci* 93:968–971.

Cappon D (1969) Orientational perception: 3. Orientational percept distortions in depersonalization. *Am J Psychiatry* 125: 1048–1056.

Cappon D, Banks R (1965) Orientational perception. II. Body perception in depersonalization. *Arch Gen Psychiatry* 13:375–379.

Cappon D, Banks R (1969) Orientational perception. IV. Time and length perception in depersonalized and derealized patients and controls under positive feedback conditions. *Am J Psychiatry* 125:1214–1217.

Cappon D, Banks R (1961) Orientational perception. *Arch Gen Psychiatry* 5:380–392.
Dening TR, Berrios GE (1989) Wilson's disease: Psychiatric symptoms in 195 cases. *Arch Gen Psychiatry* 46:1126–1134.
Dening TR, Berrios GE (1994) Autoscopic phenomena. *Br J Psychiatry* 165:808–817.
Edwards JG, Angus JW (1972) Depersonalization. *Br J Psychiatry* 120:242–244.
Fichter C, Horevitz R, Braun B (1992) Fluoxetine in depersonalization disorder. *Am J Psychiatry* 149:1750–1751.
Freeman AM III, Melges FT (1977) Depersonalization and temporal disintegration in acute mental illness. *Am J Psychiatry* 134:679–681.
Freeman AM III, Melges FT (1978) Temporal disorganization, depersonalization, and persecutory ideation in acute mental illness. *Am J Psychiatry* 135:123–124.
Hall W, Bird KD (1986) Simultaneous multiple comparison procedures for categorical data. *Aust N Z J Psychiatry* 20:350–359.
Hartmann H (1922) Ein Fall von Depersonalisation. *Zeitschrift für die gesamte Neurologie und Psychiatrie* 74:597–601.
Haug K (1936) *Die Störungen des Persönlichkeitsbewusstseins und verwandte Entfremdungserlebnisse: eine klinische un psychologische Studie*. Stuttgart: Ferdinand Enke Verlag.
Heuyer MMG, Dublineau J (1932) Syndrome de dèpersonnalisation chez un encéphalitique. *Annales Médico-Psychologiques* 90:204–207.
Janet P (1928) *De l'angoisse à la extase*. Paris: Alcan.
Krishaber M (1873) *La Névropathie Cérébro-Cardiaque*. Paris: Masson.
Lewis A (1931) The experience of time in mental disorder. *Proc R Soc Med* 25:611–620.
L'hermitte J (1939) *L'image de notre corps*. Paris: Nouvelle Revue Critique.
Mathew RJ, Wilson W, Humphreys D, Lowe J, Weithe K (1993) Depersonalization after marijuana smoking. *Biol Psychiatry* 33:431–441.
Mayer-Gross W (1935) On depersonalization. *Br J Med Psychol* 15:103–122.
Mullan S, Penfield W (1959) Illusions of comparative interpretation and emotion. *Arch Neurol Psychiatry* 81:269–284.
Reynolds HT (1984) *Analysis of nominal data* (2nd ed). London: Sage.
Roberts W (1960) Normal and abnormal depersonalisation. *J Ment Sci* 106:478–493.
Roth M (1959) The phobic anxiety-depersonalisation syndrome. *Proc R Soc Med* 52:587–595.
Saperstein JL (1949) Phenomena of depersonalization. *J Nerv Ment Dis* 110:236–251.
Schilder P (1914) *Selbstbewusstsein und Persönlichkeitbewusstsein: eine psychopathologische Studie*. Berlin: Verlag von Julius Springer.
Schilder P (1953) *Medical psychology*. New York: Wiley.
Schilder P (1935) *The image and appearance of the human body*. London: Kegan Paul.
Seidel G (1995) Suicide in the elderly in antiquity. *Int J Geriatr Psychiatry* 10:1077–1084.
Shorvon H (1946) The depersonalization syndrome. *Proc R Soc Med* 39:779–792.
Sierra M, Berrios GE (1996) A case of depersonalization. *Hist Psychiatry* 7:451–461.
Sierra M, Berrios GE (1997) Depersonalization: A conceptual history. *Hist Psychiatry* 8:213–229.
Sierra M, Berrios GE (1998) Depersonalization: neurobiological perspectives. *Biol Psychiatry* 44:898–908.
Sierra M, Berrios GE (2000) The Cambridge Depersonalization Scale: A new instrument for the measurement of depersonalization. *Psychiatry Res* 93:153–164.
Simeon D, Gross S, Guralnik O, Stein D, Schmeidler J, Hollander E (1997) Feeling unreal: 30 cases of DSM-III-R depersonalization disorder. *Am J Psychiatry* 154:1107–1113.
Winnik H (1947–1948) On the structure of the depersonalization neurosis. *Br J Med Psychol* 21:268–277.